ANDROID PROGRAMMING PRACTICAL 13

Module 2 · Threads and Handlers for background processing

Aim

Demonstrate Threads and Handlers for background processing.

Project name: ThreadHandlerDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create a UI with a Start Task button and a TextView for status.

2. Create a background Thread or Executor task for a time-consuming demonstration operation.

3. Do not update UI widgets directly from the worker thread.

4. Use a main-thread Handler or runOnUiThread() to post progress/results to the UI thread.

5. Handle task completion and avoid leaking an Activity reference after it is destroyed.

6. Run the app and verify the UI remains responsive while the task runs.

7. Test repeated taps and ensure duplicate work is controlled or cancelled appropriately.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

A background task runs without blocking the UI and returns a result to the main thread.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.